

Dynamic Cross-Sectional Factor Model with Mean-Variance Optimisation

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Abstract

This paper develops a unified quantitative equity investment framework combining a dynamic cross-sectional factor model with a statistical risk engine and constrained portfolio optimisation. Five economically motivated factors—market beta, momentum, size, low volatility, and long-term reversal—are constructed from cross-sectional equity data and transformed through winsorisation, standardisation, and sequential orthogonalisation.

Predictive signals are dynamically weighted using rolling Information Coefficients (ICs), allowing the model to adapt to changing market regimes. In parallel, a Principal Component Analysis (PCA) framework is used to estimate a low-dimensional representation of the covariance structure of returns. A residual covariance component is derived from the difference between empirical and PCA-reconstructed covariance matrices.

Portfolio weights are constructed using risk-adjusted inverse covariance scaling and further refined through mean-variance optimisation under leverage and market-neutrality constraints. The framework is evaluated through backtesting and benchmark comparison.

1 Introduction

Equity returns exhibit strong cross-sectional structure driven by systematic risk factors and behavioural anomalies. Classical factor models such as those of Fama and French [1] provide interpretable decompositions of expected returns, while modern quantitative approaches increasingly integrate statistical learning techniques and dynamic weighting schemes.

This paper proposes a modular framework that separates the investment process into three components:

1. **Alpha model:** construction and combination of cross-sectional predictive signals
2. **Risk model:** estimation of covariance structure using PCA
3. **Portfolio construction:** constrained optimisation using risk-adjusted signals

The objective is to build a robust and adaptable pipeline that improves stability in high-dimensional asset universes while preserving economic interpretability.

2 Theoretical Framework

2.1 Cross-Sectional Factor Model

Asset returns are modelled as:

$$r_{i,t} = \sum_{k=1}^K \beta_{i,k,t} f_{k,t} + \epsilon_{i,t}, \quad (1)$$

where $r_{i,t}$ is the return of asset i at time t , $\beta_{i,k,t}$ is the factor exposure, $f_{k,t}$ denotes factor returns, and $\epsilon_{i,t}$ is an idiosyncratic error term.

2.1.1 Market Beta

Market beta is estimated using a rolling Ordinary Least Squares (OLS) regression of individual asset returns on market returns. For each asset i , beta is computed over a 36-month lookback window:

$$r_{i,\tau} = \alpha_i + \beta_i r_{m,\tau} + \epsilon_{i,\tau}, \quad \tau \in \{t-35, \dots, t\}, \quad (2)$$

where τ denotes the time index within the rolling estimation window, $r_{m,\tau}$ is the return of the market portfolio at time τ , α_i is the intercept term, and β_i is the market beta of asset i estimated over the rolling 36-month window. The choice of 36 months balances the statistical precision with time-variation of this factor, as beta is slow moving and shorter windows lead to unstable beta.

The estimated coefficient $\hat{\beta}_i$ is used as the factor exposure. This rolling specification allows beta to adapt to structural changes in market sensitivity over time while maintaining statistical stability.

2.1.2 Momentum

Momentum is defined as the cumulative return over the past 12 months, excluding the most recent month:

$$\text{Mom}_{i,t} = \prod_{\tau=t-12}^{t-2} (1 + r_{i,\tau}) - 1. \quad (3)$$

The exclusion of the most recent month $t-1$ in the momentum construction is motivated by well-documented short-term reversal effects in equity returns [2]. This exclusion is justified by the presence of short-horizon mean reversion driven by several market frictions:

1. **Market microstructure effects.** Bid-ask bounce and infrequent trading induce mechanical negative autocorrelation in returns. Prices tend to oscillate around fundamental value due to discrete trading and transaction costs, causing the most recent return to be a noisy signal of true underlying drift.
2. **Investor overreaction.** Empirical studies suggest that sharp price movements are often followed by partial reversals in the subsequent period, consistent with behavioural overreaction to news or earnings announcements.
3. **Temporary price pressure.** Large institutional trades and order imbalances can temporarily displace prices away from fundamental value. As liquidity replenishes, these effects partially reverse in the following month.

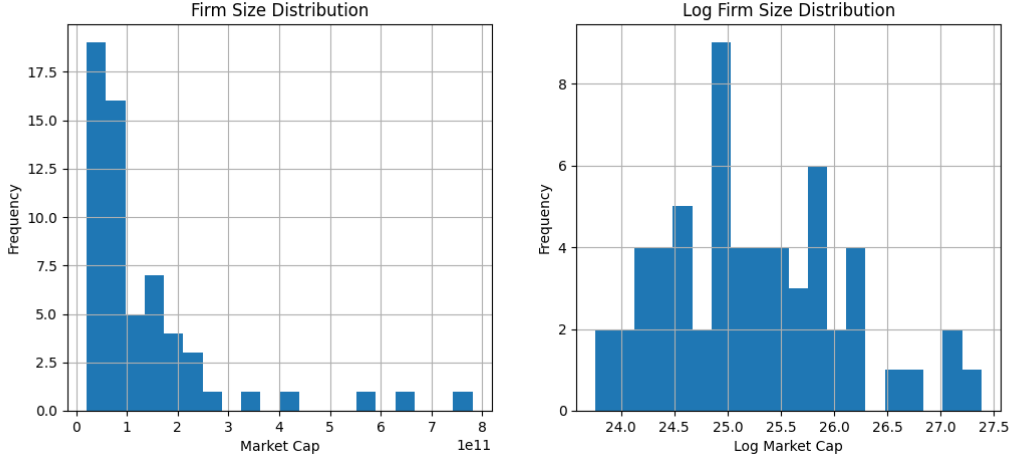


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As a result of these effects, the return at time $t - 1$ is considered predominantly noise rather than informative signal, and is therefore excluded from the momentum formation window to improve the signal-to-noise ratio and enhance predictive stability.

2.1.3 Size

Firm size is defined as the natural logarithm of market capitalisation:

$$\text{Size}_{i,t} = \log(\text{Market Cap}_{i,t}). \quad (4)$$

The log transformation follows standard Barra-style factor construction [3], reducing skewness in the cross-sectional distribution of firm sizes and preventing dominance by extreme large-cap stocks (See fig. 1). This transformation also improves numerical stability and cross-sectional comparability.

From an economic perspective, the size effect is often interpreted as compensation for liquidity risk (smaller firms have less shares outstanding and therefore are less liquid), informational inefficiency (smaller firms have less analyst coverage and forecasts), and higher uncertainty (smaller firms are more dependent on fewer key strategic partnerships, are less diversified and are more vulnerable in economic downturn) associated with smaller firms.

2.1.4 Low Volatility

The low-volatility factor is constructed from the rolling standard deviation of monthly returns over a 12-month window:

$$\text{Vol}_{i,t} = \sigma(r_{i,\tau}), \quad \tau \in [t - 12, t]. \quad (5)$$

The choice of a 12-month window is based on volatility being a highly time-varying and regime dependent signal, and low-vol strategies aim to capture near term risk characteristics. It is also to avoid outdated risk estimates.

The factor signal is then defined as the negative of realised volatility:

$$\text{Lowvol}_{i,t} = -\text{Vol}_{i,t}. \quad (6)$$

This transformation ensures that stocks with lower realised volatility receive higher factor scores.

The empirical motivation stems from the well-documented low-volatility anomaly [4], where low-risk assets outperform high-risk assets on a risk-adjusted basis. This deviation from CAPM predictions is commonly attributed to leverage constraints limiting investors' ability to amplify exposure to low-risk assets.

2.1.5 Long-Term Reversal

The long-term reversal factor captures mean reversion in equity returns over extended horizons. The signal is constructed using cumulative returns over the period $[t - 36, t - 12]$:

$$\text{LTR}_{i,t} = - \left(\prod_{\tau=t-36}^{t-12} (1 + r_{i,\tau}) - 1 \right). \quad (7)$$

This design explicitly excludes the most recent 12 months to separate long-term reversal from medium-term momentum effects.

The economic intuition behind long-term reversal includes:

1. Investor overreaction to sustained news flows, leading to price overshooting
2. Valuation mean reversion following prolonged deviations from fundamental value
3. Earnings mean reversion, where extreme underperformance is statistically followed by recovery in fundamentals

This factor complements momentum by capturing a distinct time-scale of predictability in cross-sectional returns.

2.2 Factor Signal Construction

For each factor $k \in 1, \dots, 5$ at time t , raw signals $s_{i,t}^{(k)}$ of asset $i = 1, \dots, N$ are constructed and processed in the following order:

Cross-sectional standardisation:

$$s_{i,t}^{(k)} \rightarrow \tilde{s}_{i,t}^{(k)} = \frac{s_{i,t}^{(k)} - \mu_t^{(k)}}{\sigma_t^{(k)}} \quad (8)$$

where $\mu_t^{(k)}$ is the cross-sectional mean, and $\sigma_t^{(k)}$ is the cross-sectional standard deviation.

Winsorisation:

Data points outside the 1st and 99th percentile are set to the 1st and 99th percentile. This replaces extreme outliers by replacing them with less extreme values to guard against fat tails and data errors, which may affect position sizes and OLS regressions. We do not define our boundaries at $\pm 3\sigma$ as distributions are often skewed.

Sequential orthogonalisation:

We construct orthogonalised factor signals using a sequential residualisation procedure. Market beta

is entirely reserved, and other factors are residualised against preceding signals.
i.e.

- $k = 1$: Market beta
- $k = 2$: Momentum
- $k = 3$: Size
- $k = 4$: Low volatility
- $k = 5$: Long-term reversal

For the first factor:

$$\hat{s}_{i,t}^{(1)} = \tilde{s}_{i,t}^{(1)} \quad (9)$$

For the second factor, we remove the linear projection onto the first factor:

$$\tilde{s}_{i,t}^{(2)\parallel} = \frac{\text{Cov}(\tilde{s}_t^{(2)}, \tilde{s}_t^{(1)})}{\text{Var}(\tilde{s}_t^{(1)})} \tilde{s}_{i,t}^{(1)} \quad (10)$$

$$\hat{s}_{i,t}^{(2)} = \tilde{s}_{i,t}^{(2)} - \tilde{s}_{i,t}^{(2)\parallel} \quad (11)$$

More generally, for $k \geq 2$, each factor is orthogonalised with respect to all previous factors:

$$\hat{s}_{i,t}^{(k)} = \tilde{s}_{i,t}^{(k)} - \sum_{j=1}^{k-1} \frac{\text{Cov}(\tilde{s}_t^{(k)}, \tilde{s}_t^{(j)})}{\text{Var}(\tilde{s}_t^{(j)})} \hat{s}_{i,t}^{(j)} \quad (12)$$

2.3 Information Coefficient Weighting

Predictive power is measured via the Information Coefficient:

$$IC_{i,t}^{(k)} = \text{Corr} \left(\tilde{s}_{i,t}^{(k)}, r_{i,t+1} \right) \quad (13)$$

Factor weights are defined as the rolling average of a 24-month lookback window $\bar{IC}_{i,t}^{(k)}$. This yields a factor weight for each factor k for each asset i at time t . The factor score $f_{i,t}$ for each asset is then the scalar product of the rolling average and orthogonalised signals over all factors:

$$f_{i,t} = \sum_{k=1}^5 \bar{IC}_{i,t}^{(k)} \hat{s}_{i,t}^{(k)} \quad (14)$$

3 PCA-Based Risk Model

3.1 PCA Decomposition

Let Σ_{sample} denote the sample covariance matrix of asset returns computed over a rolling window of 36 months. This is because PCA relies on a stable covariance structure and is sensitive to sampling noise. This can also be justified by a structural interpretation as market style factors are slow-evolving.

Note that the subscript t is omitted.

The covariance matrix is decomposed as:

$$\boldsymbol{\Sigma}_{\text{sample}} = \mathbf{B}\boldsymbol{\Lambda}\mathbf{B}^\top \quad (15)$$

where $\mathbf{B}$ is the eigenvector (column) matrix and $\boldsymbol{\Lambda}$ is the diagonal matrix of eigenvalues.

3.2 Low-Rank Approximation

The number of principal components (PCs), k , is selected such that

$$\frac{\sum_{i=1}^k \lambda_i}{\sum_{j=1}^N \lambda_j} \geq 0.80 \quad (16)$$

This ensures that 80% of the variance is accounted for. The PCA covariance estimator is:

$$\boldsymbol{\Sigma}_{\text{PCA}} = \mathbf{B}_k \boldsymbol{\Lambda}_k \mathbf{B}_k^\top \quad (17)$$

where $\mathbf{B}_k$ is the matrix of the first k eigenvectors, and $\boldsymbol{\Lambda}_k$ is the diagonal matrix containing the corresponding eigenvalues.

3.3 Residual Risk

Residual covariance is defined as the difference between the empirical covariance matrix and its low-rank PCA reconstruction in the window:

$$\mathbf{D} = \boldsymbol{\Sigma}_{\text{sample}} - \boldsymbol{\Sigma}_{\text{PCA}}. \quad (18)$$

This matrix captures the portion of covariance not explained by the leading principal components, and is interpreted as idiosyncratic or asset-specific co-movement structure not accounted for by common factors.

Due to, $\mathbf{D}$ is not used in full form due to noise and instability in off-diagonal elements. Instead, a diagonal approximation is employed:

$$\mathbf{D} \approx \text{diag}(\mathbf{D}), \quad (19)$$

leading to the final covariance matrix estimator:

$$\hat{\boldsymbol{\Sigma}} = \boldsymbol{\Sigma}_{\text{PCA}} + \text{diag}(\mathbf{D}) \quad (20)$$

The off-diagonal elements of $\mathbf{D}$ are highly sensitive to sampling noise in finite samples and tend to exhibit low signal-to-noise ratios. In high-dimensional settings (large N relative to time window T), empirical covariances are particularly unstable, and off-diagonal residual correlations are often indistinguishable from estimation noise.

By retaining only the diagonal, we impose the assumption that idiosyncratic returns are cross-sectionally uncorrelated, while allowing for heteroskedasticity across assets. This is consistent with standard assumptions in factor risk models, including Barra-style decompositions [5], where:

$$\text{Cov}(\epsilon_{i,t}, \epsilon_{j,t}) \approx 0 \quad \text{for } i \neq j, \quad (21)$$

but $\text{Var}(\epsilon_{i,t})$ is asset-specific and time-varying. Therefore, the approximation is not solely motivated by heteroskedasticity, but primarily by estimation stability and the high-dimensional noise structure of empirical covariance matrices.

4 Portfolio Construction

4.1 Long-Short Quantile Portfolio Construction

To evaluate the predictive power of the cross-sectional factor model, a long-short portfolio is constructed based on ranked factor scores. At each rebalancing period t , assets are sorted according to their composite factor scores $f_{i,t}$.

The investment universe is partitioned into two disjoint sets:

- Long leg: top 20% of assets ranked by $f_{i,t}$
- Short leg: bottom 20% of assets ranked by $f_{i,t}$

Let $\mathcal{L}_t$ and $\mathcal{S}_t$ denote the sets of long and short positions respectively, with equal cardinality:

$$n_L = n_S = n$$

where n corresponds to the number of assets selected in each tail (e.g. for 60 assets total, $n = 12$).

Formally:

$$\mathcal{L}_t = \{i \mid f_{i,t} \in Q_{80-100}\}, \quad \mathcal{S}_t = \{i \mid f_{i,t} \in Q_{0-20}\} \quad (22)$$

where Q_{a-b} denotes the cross-sectional quantile range at time t .

Portfolio weights are assigned uniformly across each leg. Each long position receives weight $+\frac{1}{2n}$, while each short position receives weight $-\frac{1}{2n}$:

$$w_{i,t} = \begin{cases} \frac{1}{2n}, & i \in \mathcal{L}_t \\ -\frac{1}{2n}, & i \in \mathcal{S}_t \\ 0, & \text{otherwise} \end{cases} \quad (23)$$

This construction ensures exact dollar neutrality:

$$\sum_i w_{i,t} = 0 \quad (24)$$

and unit gross exposure:

$$\sum_i |w_{i,t}| = 1 \quad (25)$$

4.2 Risk-Adjusted Allocation

Let $\mathbf{f}_t$ denote the aggregated factor score vector.

Normalisation:

$$\tilde{\mathbf{f}}_t = \frac{\mathbf{f}_t}{\sum_i f_{i,t}} \quad (26)$$

Solve for weights:

$$\mathbf{w}_t = \hat{\Sigma}^{-1} \tilde{\mathbf{f}}_t \quad (27)$$

where $\mathbf{w}_t$ is the weight vector at time t .

Portfolio constraints: Final weights are de-meaned and normalised according to (24) and (25). The Risk-Adjusted Portfolio is constructed using these sets of weights.

4.3 Mean-Variance Optimisation

A classical mean-variance optimisation problem is solved:

$$\max_{\mathbf{w}} \mathbf{w}^\top \tilde{\mathbf{f}}_t - \frac{\lambda}{2} \mathbf{w}^\top \hat{\Sigma} \mathbf{w} \quad (28)$$

The scalar parameter $\lambda > 0$ controls the trade-off between expected portfolio return and risk. A higher value of λ corresponds to greater risk aversion, penalising portfolio variance more heavily and resulting in more conservative allocations. λ may also be interpreted as a reduced-form representation of funding tightness, whereby investors with more binding financing or leverage constraints exhibit a higher effective level of risk aversion and therefore choose less risky portfolios.

In this framework, the traditional expected return vector μ is replaced by $\tilde{\mathbf{f}}_t$, which represents model-implied alpha signals derived from cross-sectional factor exposures rather than statistical return forecasts. This substitution reflects the interpretation of factor scores as relative return predictors within each rebalancing period, consistent with cross-sectional asset pricing approaches.

5 Methodology

5.1 Data

Equity monthly adjusted closing prices are sourced from `yfinance` and transformed into a clean cross-sectional return panel for factor modelling and portfolio construction.

5.1.1 Universe Construction

The investment universe consists of a diversified set of large-cap U.S. equities spanning major GICS-style sectors. For each sector, a representative subset of highly liquid constituents is selected to ensure cross-sectional dispersion and heterogeneous exposure to standard equity risk factors including value, momentum, quality, and low volatility.

The full universe is defined as:

$$\mathcal{U} = \bigcup_{s=1}^S \mathcal{U}_s, \quad (29)$$

where $\mathcal{U}_s$ denotes the set of assets within sector s .

5.1.2 Bias Considerations

Survivorship Bias

The universe is constructed using a static selection of historically liquid large-cap equities across sectors, rather than being restricted to current index constituents. This reduces bias arising from index reconstitution effects and changes in eligibility criteria over time. However, the approach does not fully eliminate survivorship bias, as the sample is implicitly conditioned on firms that remain observable in the current cross-section. Firms that exited the investable universe due to bankruptcy, delisting, or sustained underperformance are not explicitly represented, which may lead to an upward bias in historical performance estimates.

Look-ahead Bias

To mitigate look-ahead bias, all signals and portfolio weights are constructed using only information available at time t . In particular, factor exposures and financial data are lagged to ensure that accounting data and return realisations are not inadvertently incorporated into contemporaneous portfolio formation. Any use of the current investable universe is applied consistently across the backtest, without conditioning on future inclusion or exclusion events.

Selection Bias

The restriction to highly liquid large-cap equities introduces selection bias by excluding smaller or less liquid firms. While this improves tradability and reduces microstructure noise, it may overstate achievable diversification and understate exposure to illiquidity premia present in the broader equity market.

Rebalancing and Data Snooping Bias

Frequent rebalancing and iterative model refinement can introduce implicit data snooping, particularly if parameter choices are influenced by in-sample performance. To address this, the strategy parameters are fixed ex-ante, and performance is evaluated strictly out-of-sample across rolling windows.

Microstructure Bias

Using monthly returns mitigates bid-ask bounce and high-frequency noise; however, residual microstructure effects may persist for less liquid names, potentially biasing estimated covariances and factor loadings.

5.2 Pipeline

- Construct factor signals (beta, momentum, size, volatility, reversal)
- Standardise, windsorise, and orthogonalise signals
- Compute rolling IC and construct dynamic factor weights
- Estimate covariance using rolling PCA (36-month window)
- Construct risk-adjusted portfolio using inverse covariance weighting

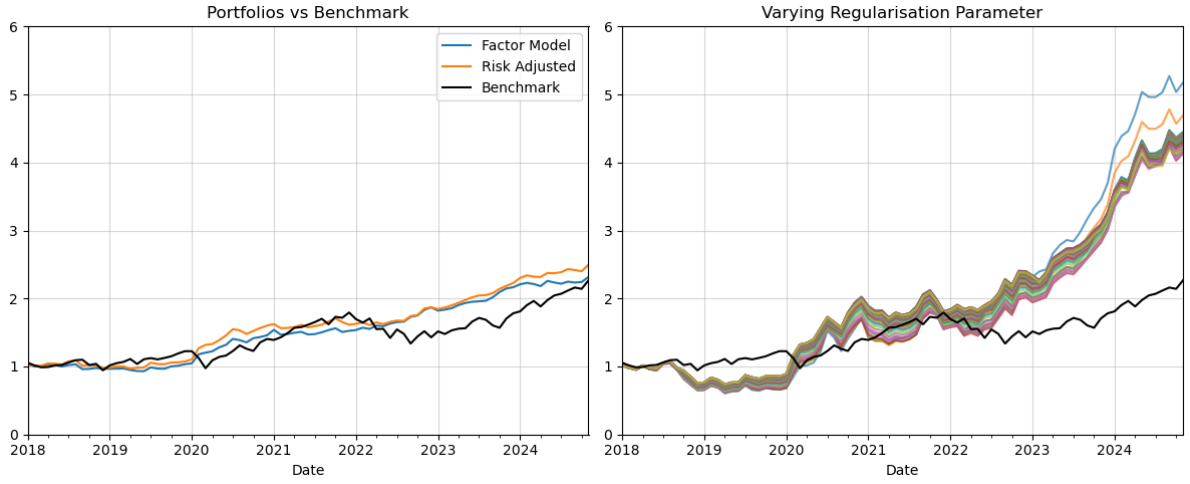


Figure 2: λ for the 2 highest performing portfolios are 1 and 2 respectively.

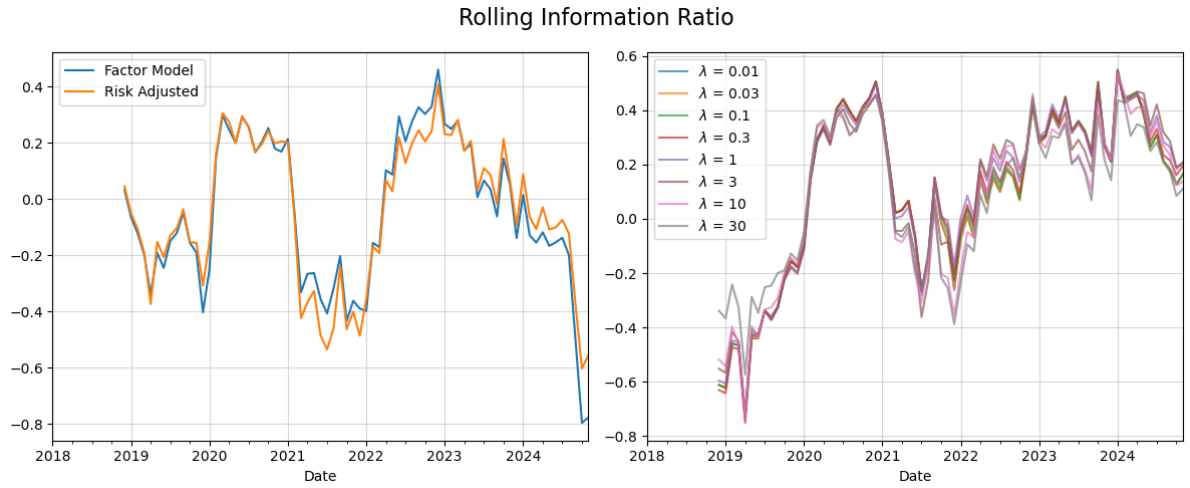


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- Solve mean-variance optimisation problem
- Evaluate performance via backtesting

6 Results

	Factor Model	Risk-Adjusted Model	Mean-Variance ($\lambda = 1$)
Cumulative PnL	149%	145%	458%
Sharpe Ratio	1.49	1.35	0.94
Max Drawdown	-9%	-13%	-41%

Table 1: Performance comparison of portfolio construction methods.

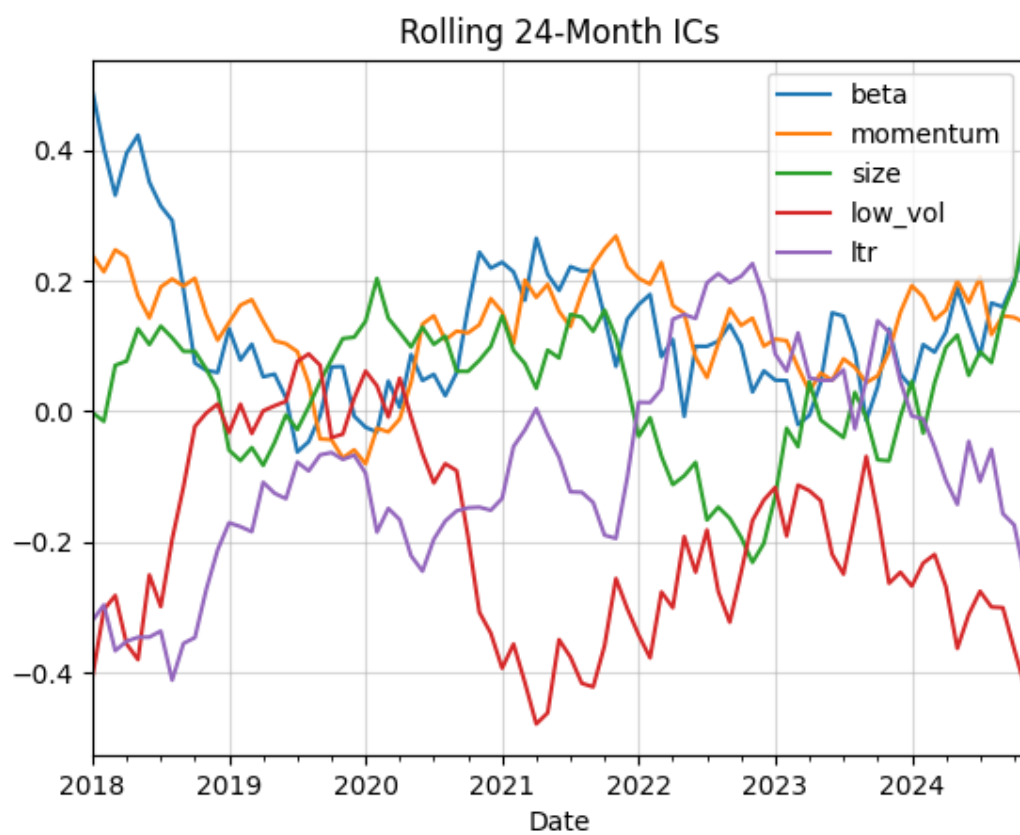


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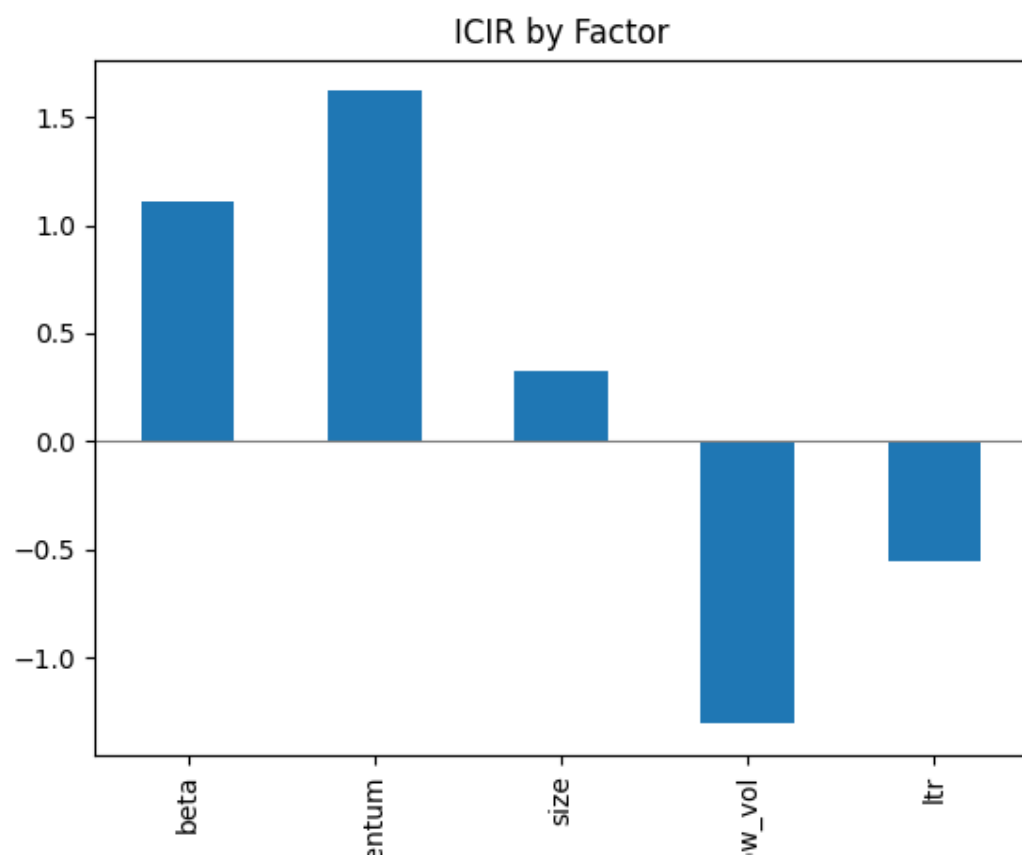


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7 Conclusion

This paper presents a modular quantitative investment framework integrating dynamic cross-sectional factor modelling, PCA-based covariance estimation, and constrained portfolio optimisation.

The separation between alpha generation and risk modelling enhances interpretability and robustness. Dynamic IC-based weighting improves adaptability across market regimes, while PCA provides a stable low-dimensional representation of covariance structure in high-dimensional asset spaces.

Future extensions include transaction cost modelling, sector neutrality constraints, and time-varying covariance models such as DCC-GARCH.

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